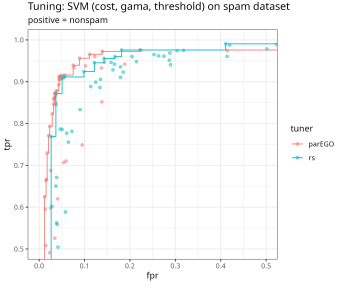
Optimization in Machine Learning

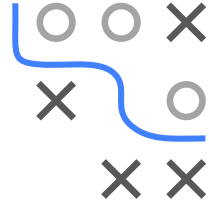
Multi-criteria Optimization
Practical Applications



Learning goals

- ROC optimization
- Efficient models
- Fairness considerations

PRACTICAL APPLICATIONS IN MACHINE LEARNING



ROC Optimization: Balance *true positive* and *false positive* rates

- Typically used in unbalanced classification with unspecified costs.
- Could also consider other ROC metrics, e.g., *positive predictive value* or *false discovery rate*.

Efficient Models: Balance *predictive performance* with *prediction time*, *energy consumption*, and/or *model size*.

- Time: production models need to predict fast.
- Size / Energy: model might be deployed on a mobile/edge device with power constraints.

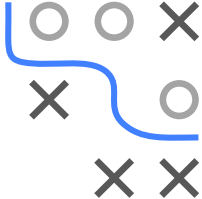
Sparse Models: Balance *predictive performance* and *number of used features*—for cost efficiency or interpretability. **Fair Models**: Balance *predictive performance* and *fairness*.

- Model should not disadvantage certain subgroups (e.g. by gender).
- Multiple approaches exist to quantify fairness.

ROC OPTIMIZATION - SETUP

Again, we want to train a *spam detector* on the popular Spam dataset.

▸ Click for source



- Learner: SVM with RBF kernel
- Hyperparameters:

`cost`	$[2^{-15}, 2^{15}]$
γ	$[2^{-15}, 2^{15}]$
Threshold t	$[0, 1]$

- Objective: *minimize* false positive rate (FPR) and *maximize* true positive rate (TPR), via 5-fold CV
- Optimizer: multi-criteria Bayesian optimization
  - ParEGO with $\rho = 0.05$, $s = 100000$
  - Acquisition: Confidence Bound ($\tau = 2.0$)
  - Budget: 100 evaluations
- Tuning is done on a training holdout, and the hyperparameter configs on the estimated Pareto front are validated on an external test set.

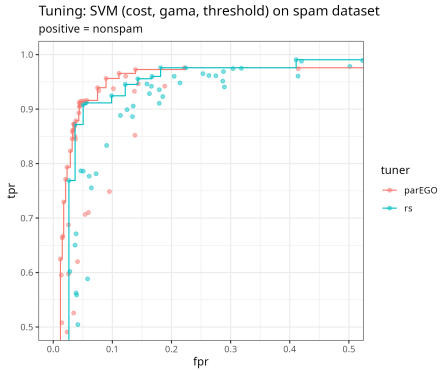
The threshold t could also be optimized post-hoc separately.

ROC OPTIMIZATION - RESULT I

- Compared to *random search*, many *ParEGO* evaluations lie on the Pareto front.
- The *ParEGO* Pareto front dominates most random-search solutions.
- Dominated hypervolume (w.r.t. reference point $(0, 1)$):

  ParEGO: 0.965

  random: 0.959



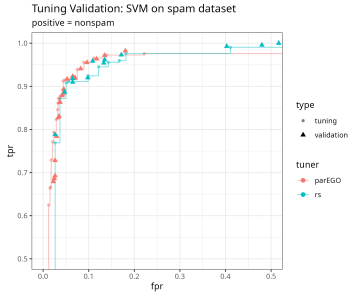
ROC OPTIMIZATION - RESULT II

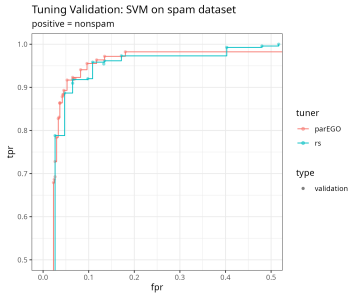
Validate configurations on the estimated Pareto front:

- The performance changes slightly on a new holdout set.
- TPR improved a bit, but FPR got slightly worse.
- Some configs become dominated once tested on the new holdout.
- Dominated hypervolume on the validation set:

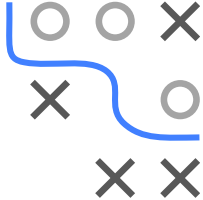
  ParEGO: 0.960

  random: 0.961



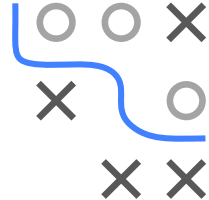


EFFICIENT MODELS - OVERVIEW



- “Efficiency” can refer to:
  - Memory consumption
  - Training or prediction time
  - Number of features used
  - Energy consumption
  - . . .
- Some hyperparameters strongly impact efficiency:
  - Number of trees (RF, GBDT)
  - Number, size, type of layers (NN)
  - L1 regularization strength
- Others might have no direct influence on efficiency.
- Typical scenario: searching over multiple algorithms with different cost–performance trade-offs.

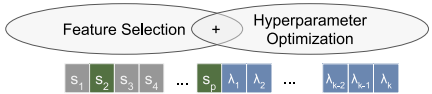
EFFICIENT MODELS - EXAMPLE: FEATURE SELECTION I



Goal: Identify an informative subset of features with minimal drop in performance vs. using all features.

$$\min_{\boldsymbol{x}\in\mathcal{S},\ s\in\{0,1\}^p} \left(\widehat{GE}\big(\mathcal{I}(\mathcal{D}, \boldsymbol{x}, s)\big),\ \frac{1}{p}\sum_{i=1}^{p} s_i \right).$$

- Typically, feature selection and hyperparameter tuning happen in separate steps.
- Alternatively, search a good subset s and hyperparams $\boldsymbol{x}$ *simultaneously*.



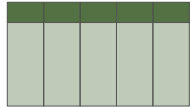
EFFICIENT MODELS - EXAMPLE: FEATURE SELECTION II

Idea: *Multi-Objective Hyperparameter Tuning and Feature Selection using Filter Ensembles* ▸ Binder et al. 2020.

- Pre-calculate multiple *feature-filter* rank vectors.

RF Feature Importance:	(2	3	4	1	5)	x	0.7
AUC:	(2	1	3	4	5)	x	0.2
Information Gain:	(1	3	5	2	4)	x	0.1

1.9 2.6 3.9 1.7 4.9



w: search space component

$$\mathrm{EF}_j(\boldsymbol{w}) = \sum_{m=1}^{M} w_m F^m(\mathcal{D})_j.$$

- New hyperparameter vector: $\boldsymbol{x} = (\tilde{\boldsymbol{x}}, w_1, \ldots, w_p, \tau)$
  - $\tilde{\boldsymbol{x}}$: learner hyperparameters
  - $(w_1, \ldots, w_p)$: weights for each filter ranking
  - $\tau \in [0, 1]$: fraction of features to keep

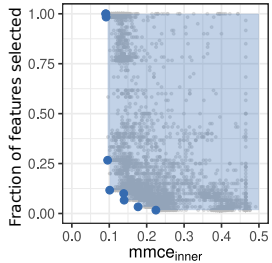
EFFICIENT MODELS - EXAMPLE: FEATURE SELECTION III

Joint FS + HP search on `Sonar` data [1]:

- Learner: SVM (RBF kernel).
- Hyperparams:

$$\begin{aligned} \texttt{cost} &\quad [2^{-10}, 2^{10}] \\ \gamma &\quad [2^{-10}, 2^{10}] \\ (w_1, \ldots, w_p) &\quad [0,1]^p \\ \tau &\quad [0,1] \end{aligned}$$

- Objective: minimize misclassification + fraction of selected features.
- Optimizer: ParEGO + random forest surrogate + LCB acquisition, 15 batch proposals, budget 2000 evals.



[1] Only tuning error is shown

EFFICIENT MODELS - EXAMPLE: FLOPS

Goal: Optimize *accuracy* vs. *FLOPS* (floating-point ops)
▸ Wang et al. 2019.

Data: CIFAR-10 (image classification).

Learner: *DenseNet* ▸ Huang et al. 2017:

- 4 dense blocks, each with multiple convolution layers that feed into each other.
- Blocks connected by convolution + max-pooling layers.

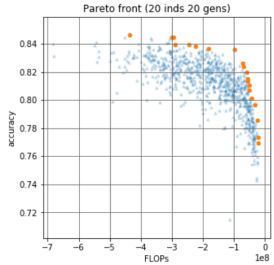
Training: 300 epochs, batch size 128, initial LR 0.1.

EFFICIENT MODELS - EXAMPLE: FLOPS (CONT.)

- Objectives: *accuracy* vs. *FLOPS per observation*.
- Search space:

`growth rate` (k)	$[8, 32]$
`layers in 1st block`	$[4, 6]$
`layers in 2nd block`	$[4, 12]$
`layers in 3rd block`	$[4, 24]$
`layers in 4th block`	$[4, 16]$

- Tuner: *Particle Swarm Optimization*, population 20, budget 400.

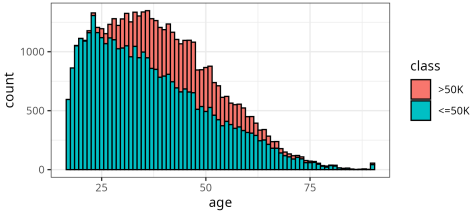


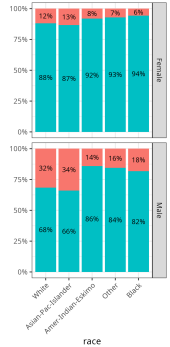
"Growth rate" = # of feature maps added each layer.

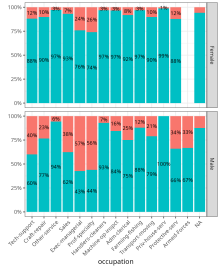
FAIR MODELS - THE `ADULT` DATASET

Dataset: `Adult`

- Source: US Census database, 1994.
  - Click for source
- 48842 observations
- Target: binary, income above 50k
- 14 features: `age`, `education`, `hours.per.week`, `marital.status`, `native.country`, `occupation`, `race`, `relationship`, `sex`, ...

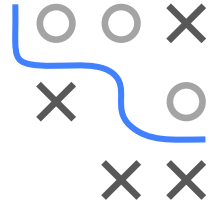






FAIR MODELS - SETUP

A toy example: build a “fair” model for predicting income (binary).



- Learner: XGBoost
- Hyperparams:

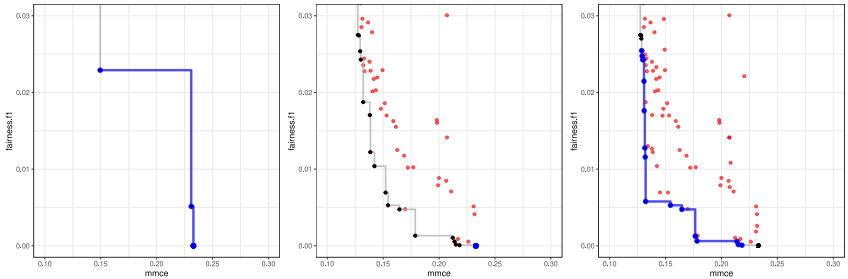
`eta`	$[0.01, 0.2]$
`gamma`	$[2^{-7}, 2^{6}]$
`max_depth`	$\{2, \ldots, 20\}$
`colsample_bytree`	$[0.5, 1]$
`colsample_bylevel`	$[0.5, 1]$
`lambda`	$[2^{-10}, 2^{10}]$
`alpha`	$[2^{-10}, 2^{10}]$
`subsample`	$[0.5, 1]$

- Objective: *misclassification error* vs. *unfairness*.
- We define “unfairness” as the absolute difference in F1-scores between male/female sub-populations:

$$L_{\text{fair}} := \left| \mathrm{F1}(y_f, \hat{f}(\boldsymbol{x}_f)) \; - \; \mathrm{F1}(y_m, \hat{f}(\boldsymbol{x}_m)) \right|.$$

Note: This is a simplistic example. Real fairness questions often need more careful definitions and domain knowledge.

FAIR MODELS - RESULTS



- Optimizer: ParEGO + random forest surrogate, restricting $[w_i]$ to $[0.1, 0.9]$ if we only care about moderate fairness/performance ranges.
- In this example, hyperparams do indeed influence the fairness measure.
- But in practice, changing standard HPs often *won't* suffice to ensure fairness.